

Real-Time Financial Fraud Detection Pipeline using Machine Learning and Stream Processing

Vinay Kumar Simma

Pradeep Yarlagadda(yarla3p)

Praneeth Reddy Ambati(ambat3p)

This project presents an end-to-end, scalable machine learning pipeline designed for real-time financial fraud detection. Traditional fraud detection systems often struggle with processing massive, high-velocity transaction volumes without severe latency. To overcome this limitation, our system integrates an optimized XGBoost classification model with a robust streaming architecture powered by Apache Kafka. The pipeline architecture consists of a simulated data producer that publishes raw transactions to a local Kafka broker, and a consumer that continuously pulls from the stream by querying a FastAPI-hosted machine learning model for immediate fraud probability evaluation. The results are securely persisted in a database and visualized through a real-time Streamlit KPI dashboard. Designed for robust cloud deployment on AWS EC2 using Docker containerization, this project successfully bridges the gap between static machine learning models and dynamic, production-ready stream processing, ensuring high accuracy and minimal latency during continuous transaction evaluation.